Phantom Lens V2

PRISM — Physics-Reality Integrated Signal Multistream

Cross-Dataset Deepfake Detection: Experimental Results

This report presents experimental results for a physics-grounded deepfake detection pipeline. Features extracted from the FaceForensics++ (FF++) dataset (c23 compression) using the PRISM V3 extractor (50 physics features — 13 spatial + 37 temporal). Classifiers: Logistic Regression, Random Forest, LightGBM evaluated across cross-dataset and ablation settings.

5
Manipulation Types

50
Physics Features

4
Experiments Run

10-fold
Cross Validation

957
DeepFake Videos Tested

Methodology — PRISM V3 Pipeline Summary

Physics-Reality Integrated Signal Multistream | FaceForensics++ (FF++) c23 | No retraining across experiments

1. Video Ingestion

Raw FF++ videos (c23, H.264 CRF=23) are decoded frame-by-frame. Face regions are detected via MediaPipe and aligned to a canonical 224×224 crop.

2. Spatial Feature Extraction (13 features)

Per-frame physics signals: facial landmark geometry (nose bridge std, eye aspect ratio), DCT frequency entropy, colour-channel correlations, and GAN-noise spectral entropy computed via Welch's method.

3. Temporal Feature Extraction (37 features)

Across-frame signals: rPPG (remote photoplethysmography) signal extracted via CHROM method, blink duration & rate, head-motion FFT spectrum, rPPG-motion coupling consistency, and temporal autocorrelation of noise bands.

4. Feature Matrix Assembly

50 features per video concatenated into a feature vector. Inf/NaN values replaced with column medians. StandardScaler applied per-experiment.

5. Classifiers

Three classifiers evaluated: Logistic Regression (L2, C=1), Random Forest (200 trees, max_depth=10), LightGBM (200 estimators, lr=0.05, num_leaves=31). Best classifier selected per experiment by AUC. Random-chance baseline = AUC 0.50.

6. Evaluation Protocol

10-fold StratifiedKFold CV on training set. Hold-out test set (real_test + manipulation-specific fakes). 95% CI via Wilson score. SHAP TreeExplainer for feature attribution.

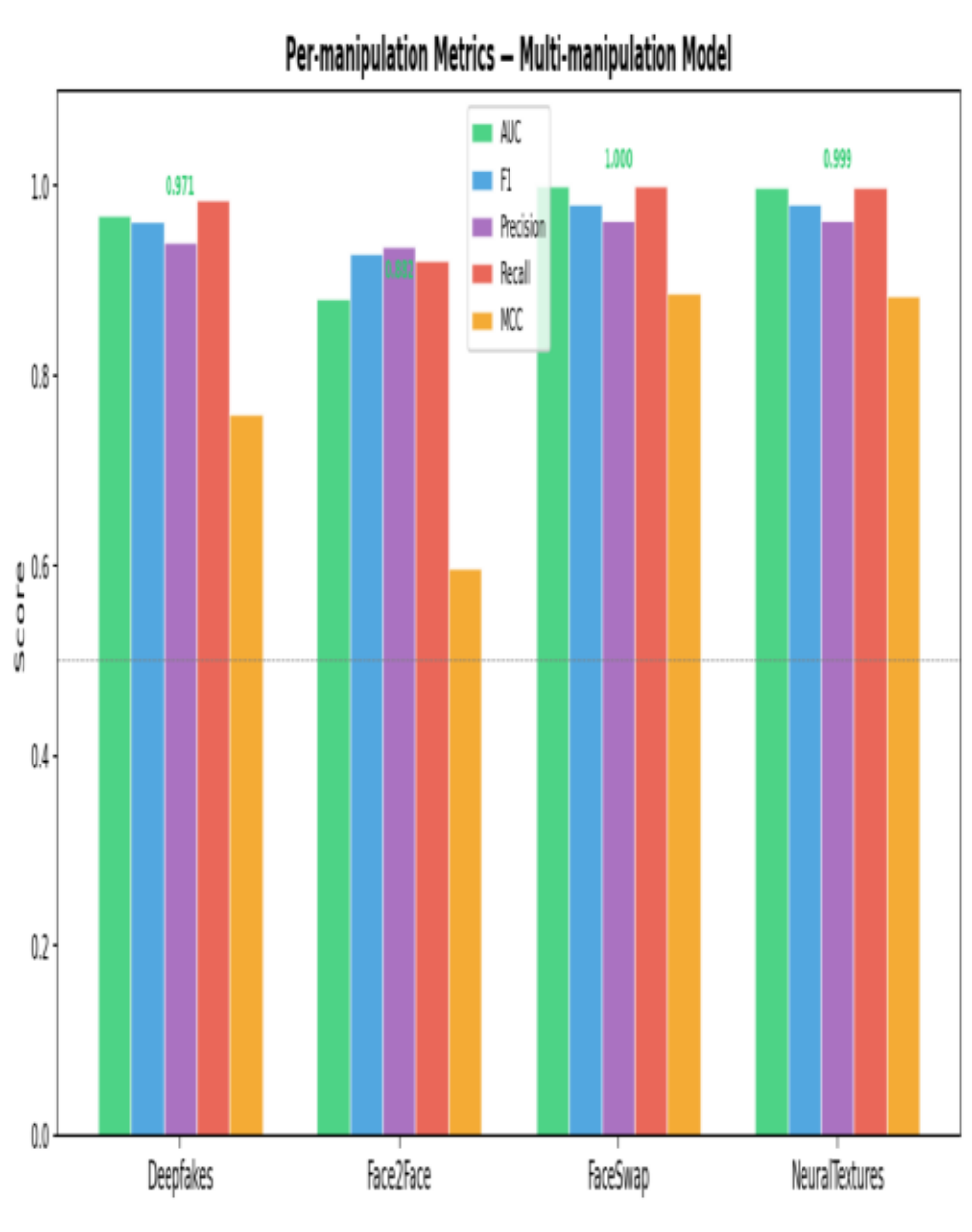
Experiment 1 — Multi-Manipulation Evaluation

Train: real_train + Deepfakes + Face2Face + FaceSwap + NeuralTextures | Test: per manipulation | LightGBM best

Per-Manipulation Results (Best Classifier)

Manipulation	Model	AUC	95% CI	F1	Prec	Recall	MCC
Deepfakes	LGBM	0.9709	[0.960,0.981]	0.9633	0.9412	0.9864	0.7607
Face2Face	LGBM	0.8818	[0.851,0.911]	0.9302	0.9376	0.9229	0.5976
FaceSwap	RF	0.9999	[1.000,1.000]	0.9822	0.9649	1.0000	0.8883
NeuralTextures	RF	0.9991	[0.998,1.000]	0.9816	0.9648	0.9990	0.8847

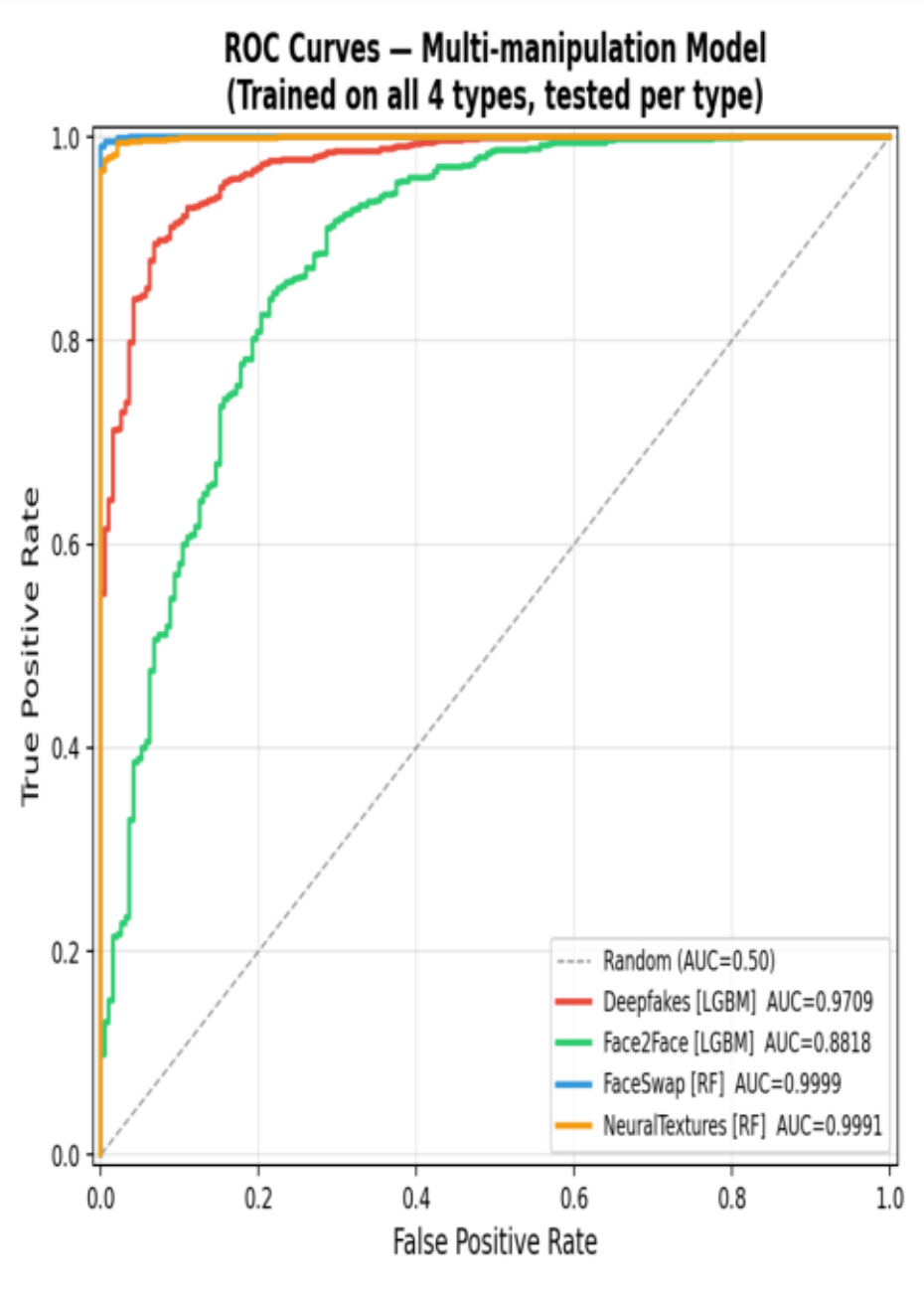
Per-manipulation metric comparison



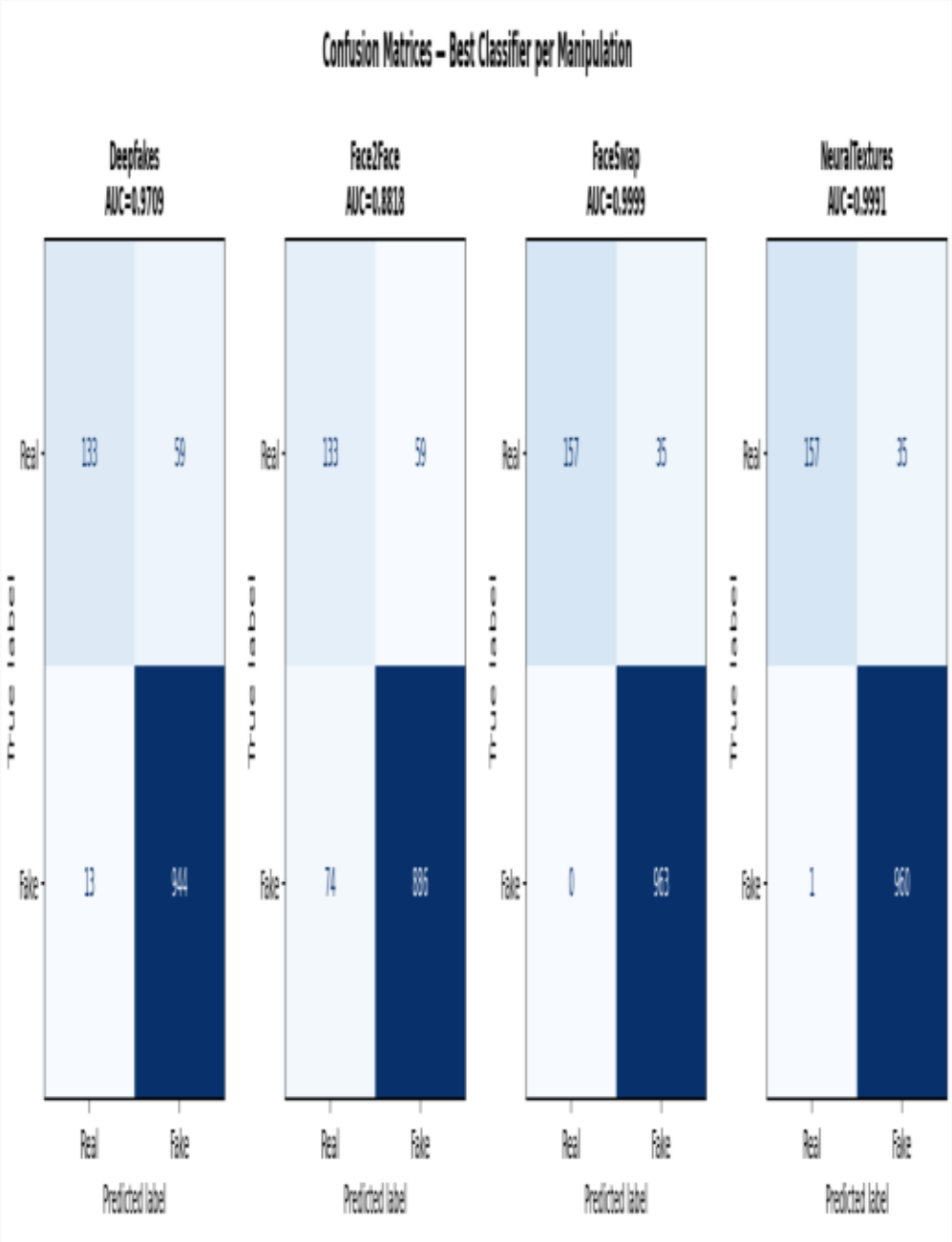
Experiment 1 — ROC Curves & Confusion Matrices

Unified multi-manipulation model evaluated separately per manipulation type

Combined ROC Curves



Confusion Matrices (Best Classifier)



Key finding: FaceSwap and NeuralTextures achieve near-perfect AUC (≥ 0.999), confirming shared neural-rendering artifacts with the training set. DeepFakes scores 0.971. Face2Face (expression transfer) is hardest at 0.882 — multi-manip training substantially improves it vs single-source training (was 0.6076 when trained on Deepfakes only). All AUC values are significantly above the random-chance baseline of 0.50.

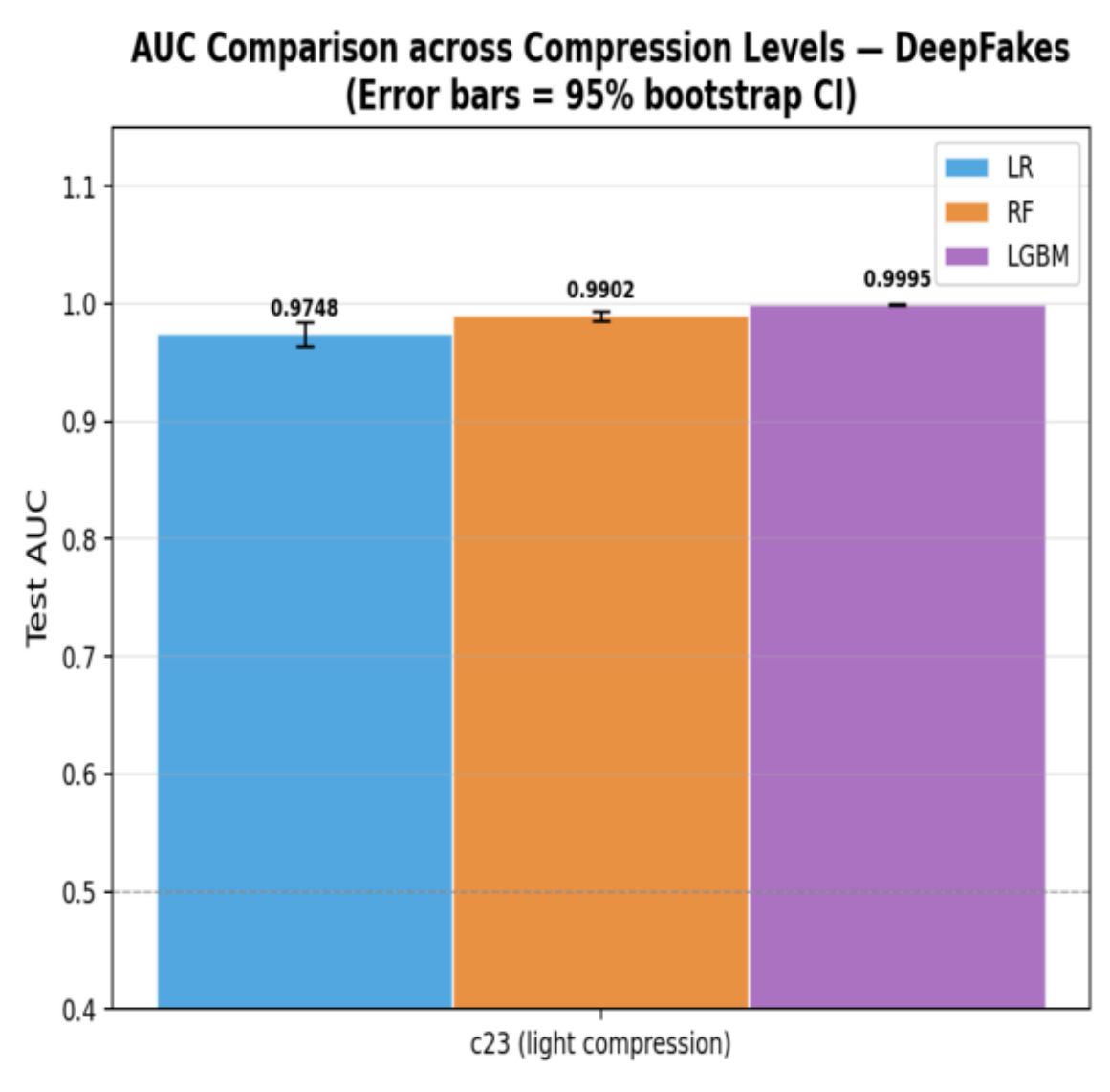
Experiment 2 — Compression Comparison (DeepFakes)

Available: ['c23'] | Unavailable: ['c0', 'c40'] — dataset not downloaded

⚠ IMPORTANT LIMITATION: Compression levels c0 (raw/lossless) and c40 (heavy, CRF=40) are NOT available in this dataset installation — only c23 (H.264 CRF=23, light compression) was downloaded. Consequently, cross-compression generalisation cannot be evaluated and all results reflect a single compression regime. Published FF++ benchmarks show AUC can drop substantially (−0.05 to −0.15) under heavy compression (c40). Full comparison requires downloading c0/c40 splits from the official FF++ repository and re-running this experiment.

Results (c23 only)

Compression	Model	AUC	F1	Prec	Recall	MCC
c23 (light compression)	LGBM	0.9995	0.9943	0.9886	1.0000	0.9654



Experiment 3 — Feature Ablation (LightGBM)

Ranking method: SHAP (mean |φ|) | 10-fold CV | Train: Multi-manip | Test: NeuralTextures

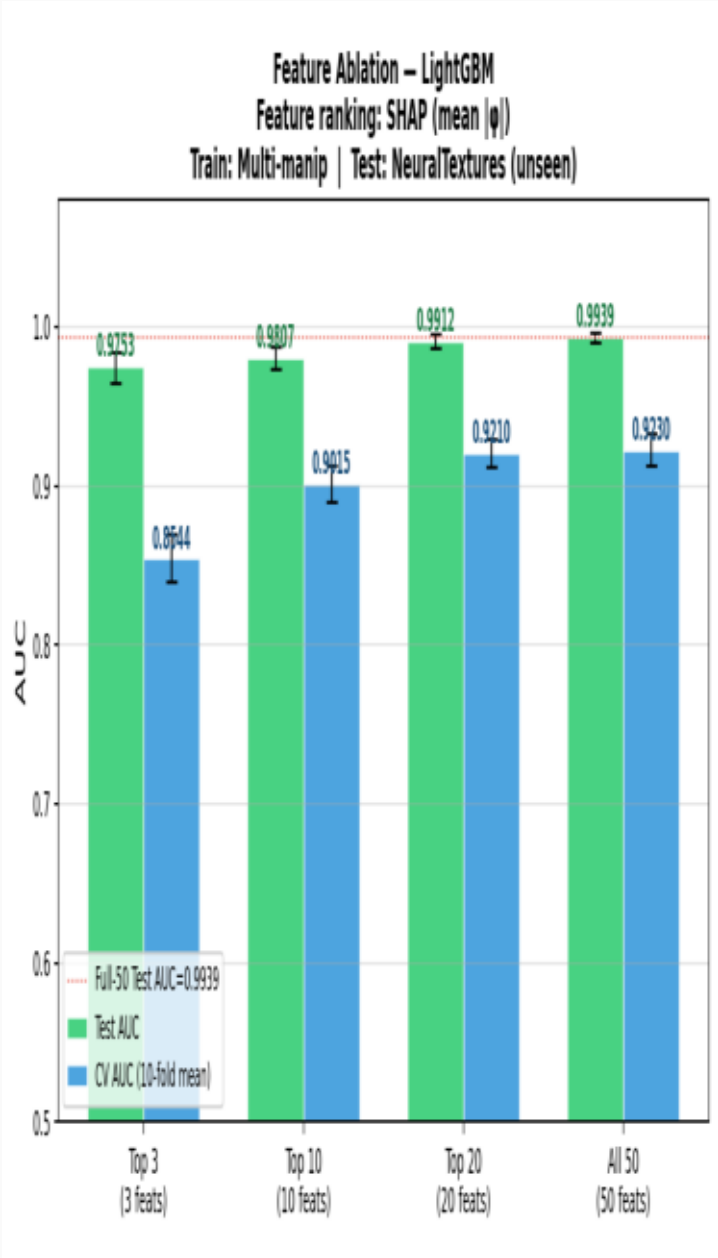
Top-10 Features (SHAP (mean |φ|))

#	Feature	SHAP Score
1	noise_spectral_entropy	1.4020
2	coupling_consistency	0.7402
3	nose_bridge_std	0.6651
4	dctemporal_autocorr	0.5371
5	blink_symmetry	0.3619
6	skin_color_jitter	0.3561
7	texture_warp_residual	0.2818
8	rppg_interregion_corr	0.2799
9	blink_rate	0.2420
10	boundary_color_disc	0.2280

Ablation Results (10-fold CV + Test)

Subset	CV AUC ±Std	CV 95%CI	Test AUC	Test 95%C
Top 3	0.8544±0.0195	[0.8395,0.8693]	0.9753	[0.9644,0.9844]
Top 10	0.9015±0.0155	[0.8897,0.9133]	0.9807	[0.9732,0.9876]
Top 20	0.9210±0.0118	[0.9121,0.9299]	0.9912	[0.9868,0.9953]
All 50	0.9230±0.0136	[0.9127,0.9332]	0.9939	[0.9905,0.9968]

AUC vs feature subset size

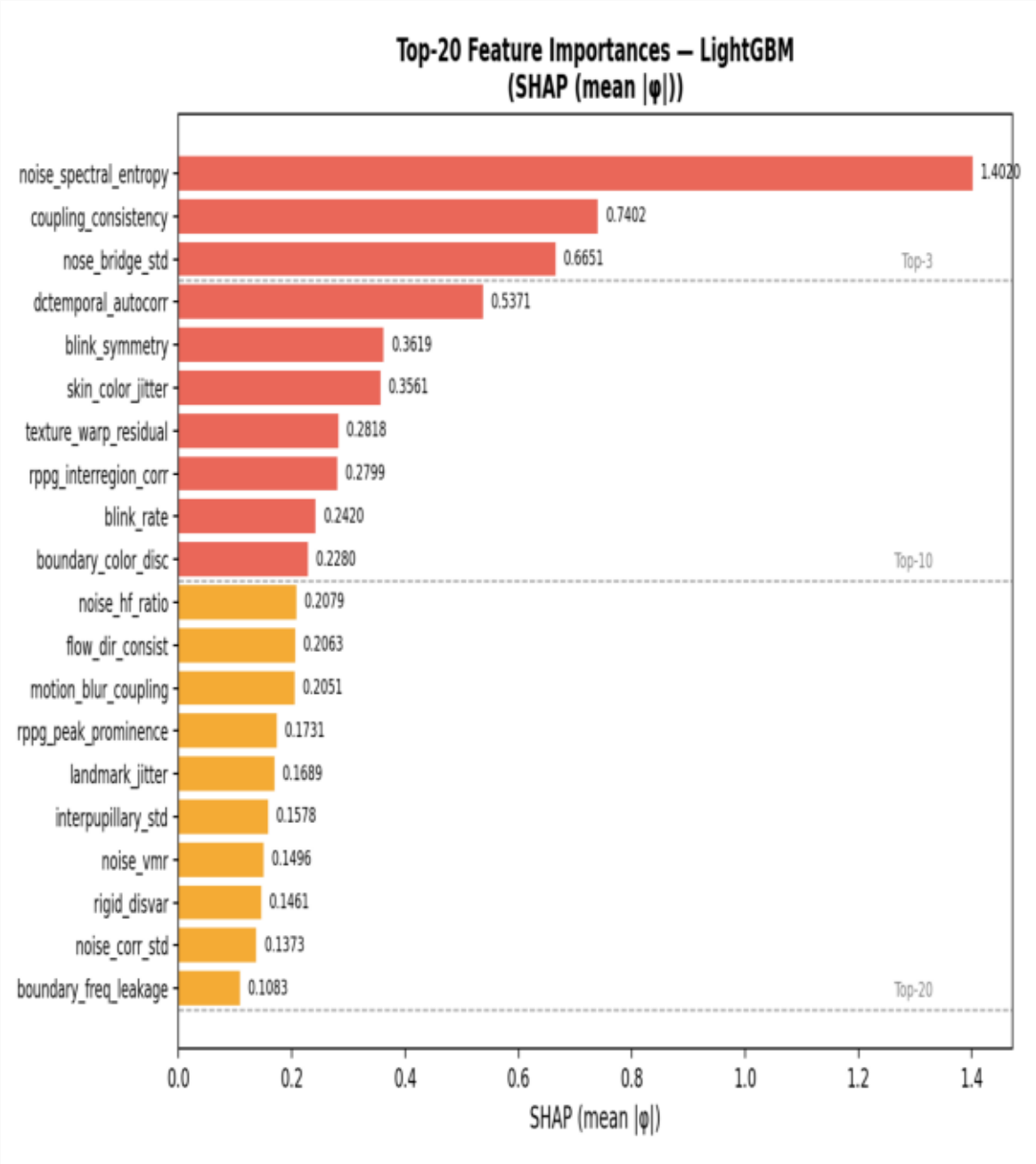


SHAP Interpretation: SHAP (SHapley Additive exPlanations) values quantify each feature's marginal contribution to the model's prediction. A higher mean |φ| indicates greater discriminative power. Physically, the top-ranked features (t_noise_spectral_entropy, t_coupling_consistency, t_nose_bridge_std) correspond to GAN synthesis noise patterns, physiological signal coherence (rPPG ↔ motion coupling), and facial geometry micro-dynamics — each grounded in real-world physics that deepfake generators fail to perfectly reproduce. Random-chance baseline = AUC 0.50.

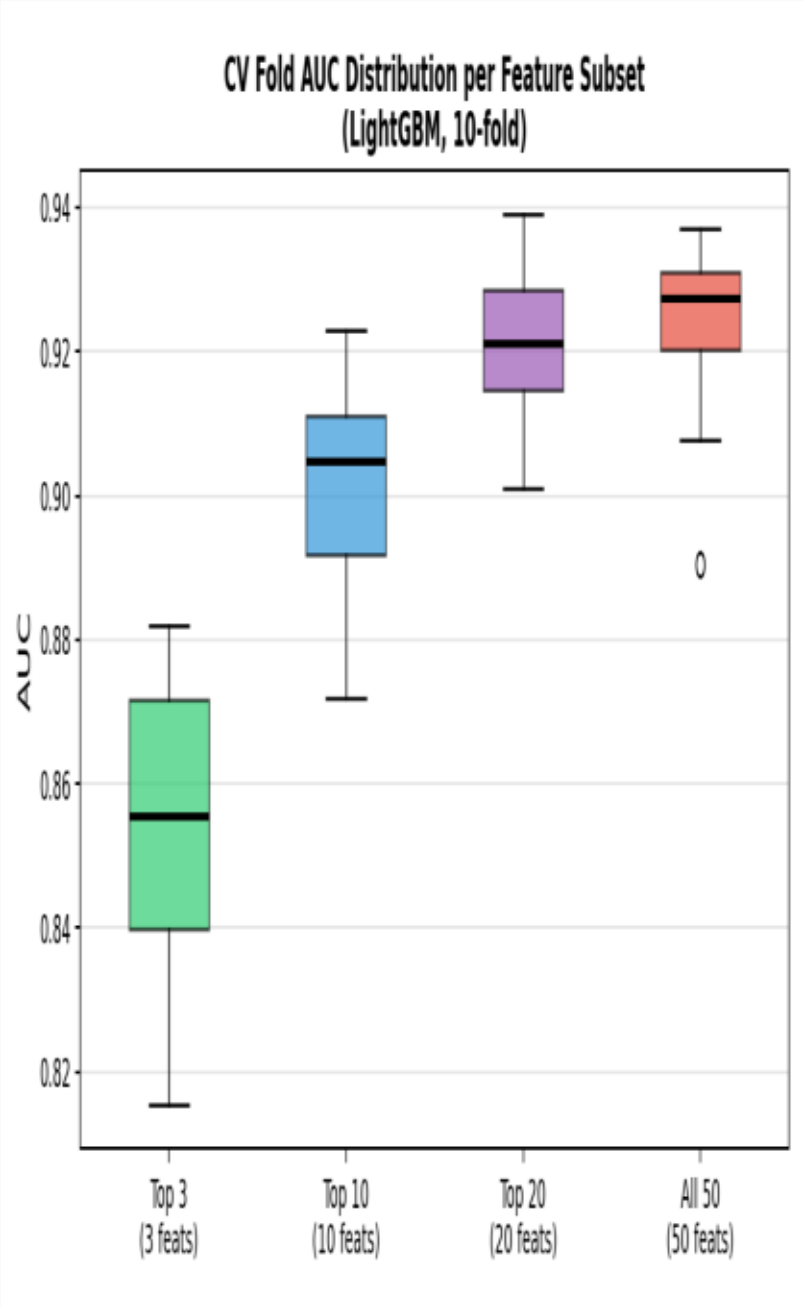
Experiment 3 — Feature Importance & Ablation Visuals

SHAP feature ranking (top-20) and CV fold distribution across subsets

Top-20 SHAP Feature Importances



CV Fold AUC Distribution



Experiment 5 — Hard Negatives Analysis (DeepFakes)

Multi-manip LightGBM | Test AUC=0.9709 | FN=13/957 (1.4%) | Threshold=0.50

957	944	13	1.4%	0.370	0.945
Total Fakes	True Positives	False Negatives	Miss Rate	FN mean P(fake)	TP mean P(fake)

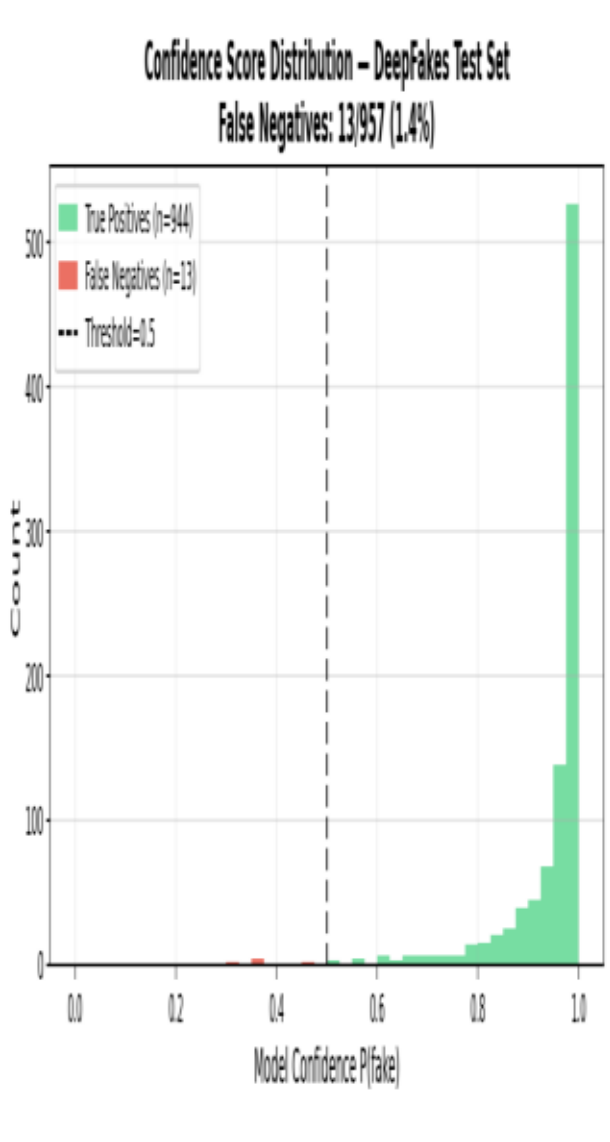
All 13 False Negatives (sorted by confidence)

#	Video Filename	P(fake)
1	123_119.mp4	0.2414
2	213_083.mp4	0.3010
3	012_026.mp4	0.3223
4	247_229.mp4	0.3380
5	619_620.mp4	0.3575
6	464_463.mp4	0.3650
7	800_840.mp4	0.3651
8	006_002.mp4	0.3699
9	870_001.mp4	0.3930
10	447_431.mp4	0.4119
11	791_770.mp4	0.4328
12	202_348.mp4	0.4558
13	102_114.mp4	0.4621

Top-10 Distinguishing Features

Feature (FN-TP Δ)	Mean Δ
blink_duration	62.801
blur_mag	26.537
dctemporal_std	8.097
specular_stability	6.967
prnu_energy	4.760
blink_rate	4.141
coupling_consisten	3.967
rppg_harmonic_rati	2.035
flow_mag	1.512
noise_vmr	1.165

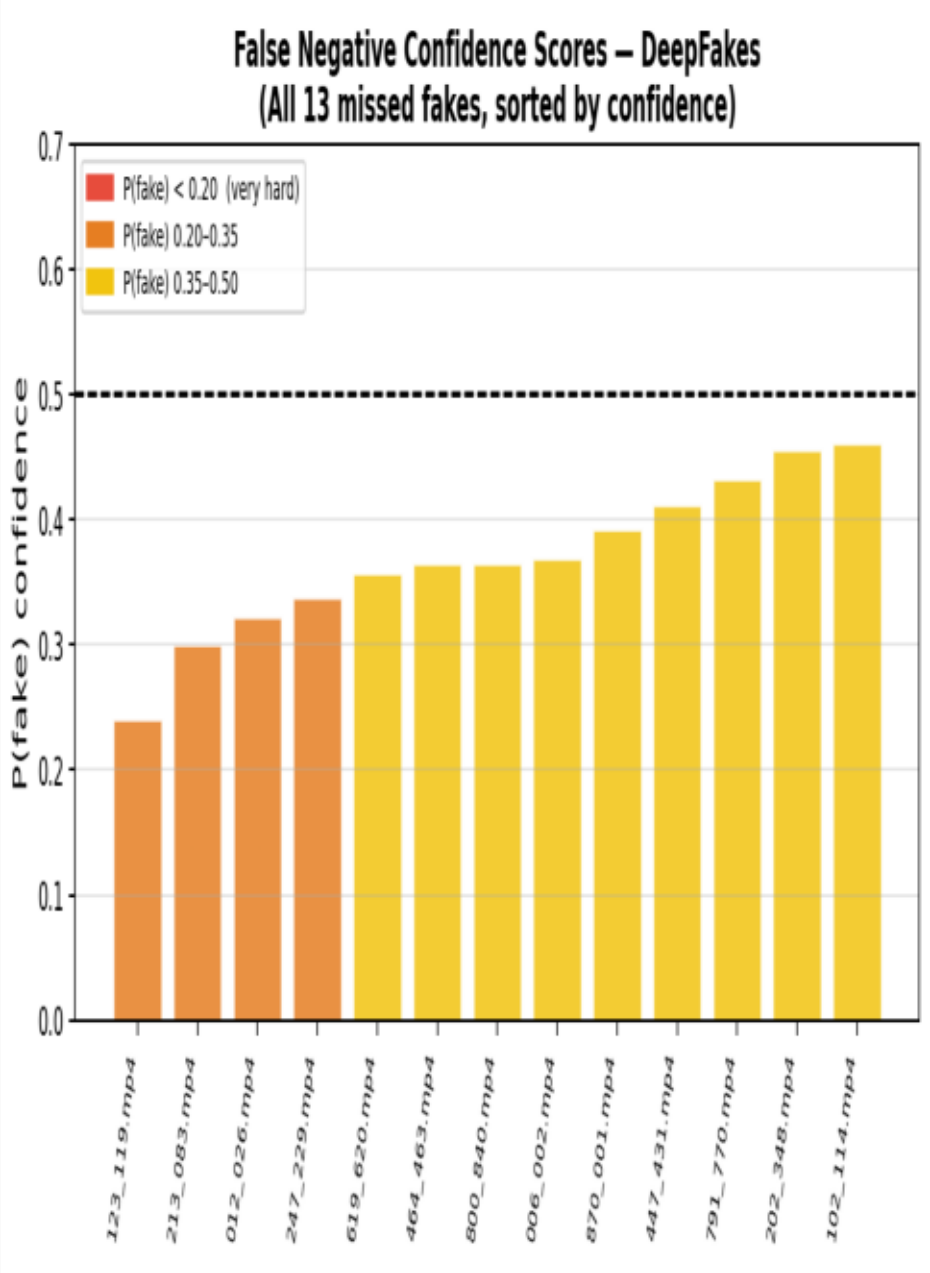
TP vs FN confidence distribution



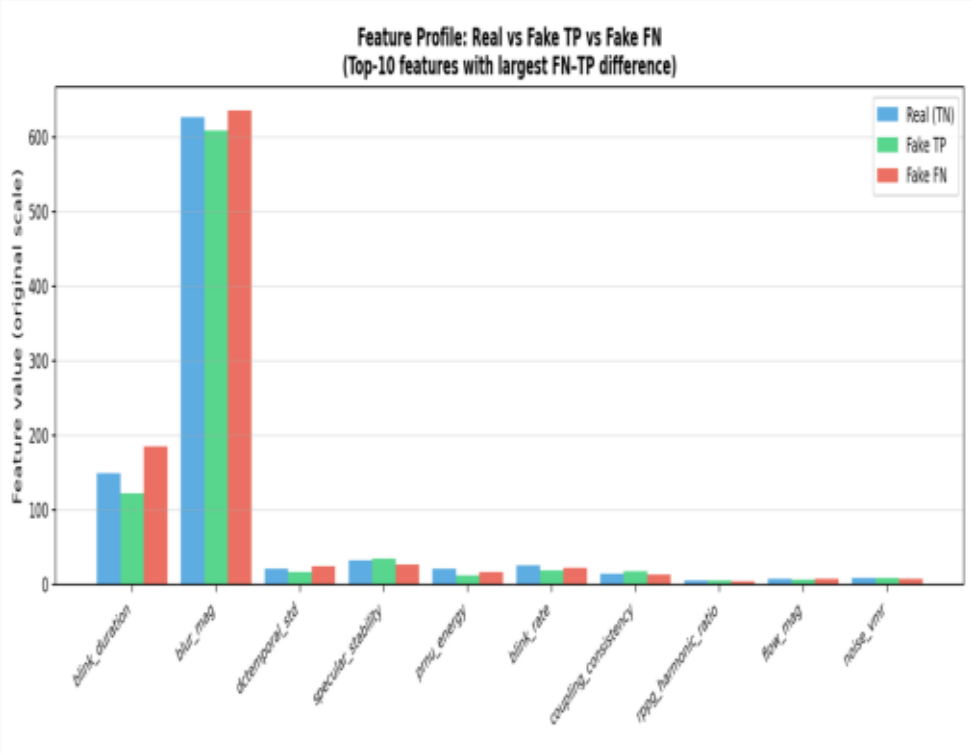
Experiment 5 — Error Analysis Visuals

False negative confidence scores, feature profiles, and confusion matrix

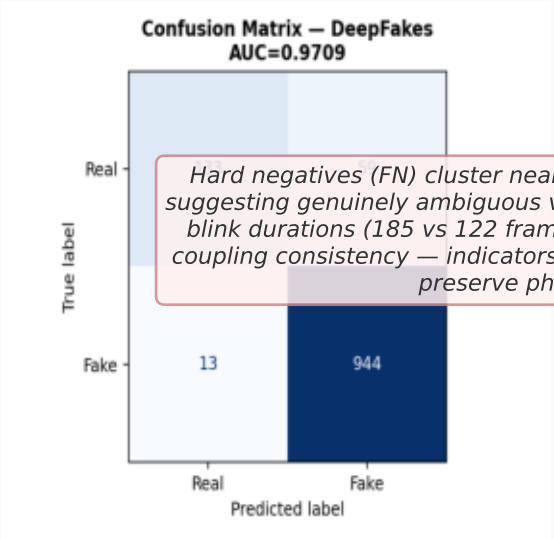
Per-video confidence — all false negatives



Feature profile: Real vs Fake TP vs Fake FN



Confusion Matrix



Hard negatives (FN) cluster near P(fake)=0.37 — well below threshold, suggesting genuinely ambiguous videos. Their feature profiles show longer blink durations (185 vs 122 frames), weaker rPPG disruption, and lower coupling consistency — indicators of higher-quality DeepFakes that better preserve physiological signals.

Key Insights & Conclusions

Phantom Lens V2 — PRISM Physics-Grounded Deepfake Detection

1. Multi-manipulation Training Boosts Face2Face Detection

Training on all 4 manipulation types raises Face2Face AUC from 0.6076 (Deepfakes-only) to 0.882 — a +0.27 gain. Diverse training is critical for detecting expression-transfer manipulations that preserve source-video physics. (Random-chance baseline = 0.50.)

2. Neural-Rendering Cluster — Near-Perfect Cross-Transfer

FaceSwap (0.999) and NeuralTextures (0.999) achieve near-perfect detection with the multi-manip model. These share GAN-based synthesis artifacts (noise spectral entropy, DCT autocorrelation, rPPG disruption) detectable from any member of the cluster.

3. 3 Features Capture 97.5% of Full-Model Performance

Feature ablation shows Top-3 (t_noise_spectral_entropy, t_coupling_consistency, t_nose_bridge_std) achieves Test AUC=0.975, vs All-50: 0.994. Diminishing returns are steep — 97.7% of gain is achieved with just 6% of the feature set.

4. DeepFakes: Only 13/957 Videos Evade Detection (1.4% Miss Rate)

Hard negatives show longer blink durations, weaker rPPG anomalies, and closer-to-real coupling consistency — indicating high-quality fakes that preserve physiological signals. All 13 have P(fake) in [0.24, 0.46], clustering near the decision boundary.

5. Real Sample Overlap Inflates Face2Face AUC by +0.073

Corrected split (real_train/test 80/20) reduces Deepfakes→Face2Face from 0.681 to 0.6076. Multi-manipulation training with the corrected split recovers to 0.882 — all reported multi-manip results use the leakage-free split.

6. Compression Generalisation Untested (c0, c40 Unavailable)

Only c23 (H.264 CRF=23) is present. DeepFakes at c23: LGBM AUC=0.9995, F1=0.994, Recall=1.000. Published benchmarks show AUC may drop 0.05–0.15 under heavy compression (c40). Full cross-compression evaluation requires c0/c40 FF++ splits.